

Machine learning based state-of-charge prediction of electrochemical green hydrogen production: *Zink-Zwischenschritt-Elektrolyseur* (ZZE)

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HIGHLIGHTS

- A novel technique for hybrid energy storage and hydrogen production is introduced.
- Various machine learning models are studied for effectiveness in SOC prediction.
- SOC prediction is crucial for future management software of ZZE systems.
- LSTM neural network showed best performance for SOC-prediction with 3.21 % MAE.

ARTICLE INFO

Keywords:

Electrolyzer
Battery
ZZE
Energy storage
State of charge
Machine learning
Sequence-to-sequence model

ABSTRACT

The intermittency of renewable energy is a key limiting factor for the successful decarbonization of both energy producing and consuming sectors. Green hydrogen has the potential to act as the central energy vector connecting hard-to-abate sectors to renewable power. However, combining energy storage and conversion for a holistic electrolyzer system remains challenging. Here, we show the innovative *Zink-Zwischenschritt Elektrolyseur* (ZZE), or Zinc Intermediate step Electrolyzer in English, that temporarily decouples the water splitting reaction and uses zinc to store electrical energy in chemical form. To perform optimal operation of a ZZE system, machine learning models were applied to predict the state of charge of a lab scale ZZE system. Using various models, we were able to determine the effectiveness of the prediction and contrast it to state of charge predictions of other energy storage systems. We show that a bi-directional long short-term memory neural network approach has the lowest error within the testing environment. This work serves to perform further ZZE development as well as state of charge prediction for other novel energy storage technologies.

Introduction

Renewables are being added to power grids at increasing rates across the globe. This has caused a number of issues related to system strength and inertia due to the intermittent nature of renewable energy sources [1]. Operational flexibility and balancing—traditionally provided by oil- or gas-powered facilities—are becoming increasingly important reliability services that are needed to balance the intermittency of renewable energy. In order to achieve deep decarbonization, this grid based balancing act will need to be redefined from balancing the daily load (electricity demand) with nimble fossil power, to balancing intermittent renewables with clean balancing options. An efficient green grid either needs to be able to efficiently connect the supply of renewable

energy to centers of electricity demand in real time or be able to store excess renewable energy to avoid curtailment. In practice, upgrading or replacing transmission lines can often prove costly [2]. And, even if demand and supply are perfectly connected, the temporal separation of peak renewable generation and peak electricity demand would still create frequency problems for the grid.

One example of this can be found in the state of California (USA). The California Independent System Operator (CAISO) boasts an exemplary high ratio of renewable energy in their energy mix and makes their grid data available publicly. According to the Electric Power Research Institute (EPRI), the CAISO duck curve, which has been historically used to describe the timing imbalance between peak demand and PV generation [3] (seen in Fig. 1), now looks like a canyon. This means that the

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<https://doi.org/10.1016/j.egyai.2024.100355>

Available online 28 February 2024

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midday trend that was present in the duck curve is becoming more intense (deeper canyon and steeper walls). Recently, CAISO's net load has reached zero or even gone negative (along with the electricity prices) during the middle of the day, meaning that in order to maintain grid stability CAISO must curtail some of the daytime renewable generation or fall back on dispatchable fossil fuel generation to correct the imbalance [4].

This concept is hardly specific to California and has been widely adopted worldwide [4]. According to Agora Energiewende, electricity prices in Germany were negative as early as in the year 2012 [5]. The accepted consensus is that negative electricity prices are a result of a surplus of renewable generation [5]. However, renewable energy is not the entire story; the negative electricity prices are also caused by a lack of flexibility within the electricity system [5]. While this certainly presents an issue, it also presents an opportunity to "fill in the canyon" with green technologies that can prevent additional flexible fossil fuel deployment by transferring what would otherwise be curtailed renewable energy to when and where it is needed, including end uses that currently fall outside the grid.

Energy storage systems (ESSs) have proved effective in mitigating grid stability problems [1]; they can be deployed modularly as a pinpoint solution to congested grids, bolstering transmission system strength and inertia [2]. The idea is relatively straight forward; allow energy to flow out of the grid to storage when it is becoming congested and back in when the frequency falls as night-time demand outpaces day-time renewable peaks. With the increasing percentage of energy production from renewables, and total demand for electric power steadily increasing, the daily temporal imbalance between generation and consumption is becoming even more extreme (duck to canyon). Energy storage technology can distribute energy peaks and modulate frequency, smooth fluctuations, and help output high-quality electrical energy [6]. However, the large-scale deployment of intermittent renewable energy with multiple time scales is not only causing operational problems but also profoundly affecting the complexity of the power system [6]. Added complexity and uncertainty can be difficult to handle for human designers using conventional methods [7].

Within the last years, the implementation of machine learning (ML) and artificial intelligence (AI)-based algorithms have steadily revolutionized various domains, offering unprecedented opportunities for advancements. This combination of novel methods is arising to work as an innovative and beneficial approach within energy research and has ranged from electrical grid stabilization by grid harmonic distortion [8] to battery electric vehicle charging forecasting [9] to state of charge (SOC) determination of batteries [10]. Being able to predict the SOC in

batteries is a fundamental aspect governing their performance, lifespan, and overall efficiency [10]. This will be discussed in more detail in the following sections.

It is important to note that according to the International Energy Agency (IEA) power generation accounts for only roughly 41 % of global greenhouse gas emissions [11]. This means that in order to achieve total decarbonization sectors such as transportation, industry, and buildings will also need to switch to renewable based power. One of the most discussed energy vectors capable of serving these sectors is green hydrogen. Traditionally, green hydrogen is only produced via electrolysis that consumes renewable power and produces hydrogen concurrently. The relevant sectors need to be decarbonized beyond the middle of the day when cheap renewable power is available. One (not so) simple solution would be to combine energy storage and electrolysis to flatten the midday renewable peak into stable green hydrogen production.

Currently, this technology combination can take one of two forms: an electrolyzer plus a hydrogen storage tank or a battery plus an electrolyzer. The former is not often seriously discussed for industrial scale projects due to the effect that this operation would have on the electrolyzer: In order to capture peaks of renewable energy the electrolyzer (and compressor) would have to have enough capacity to capture the maximum power output of the renewables. The electrolyzer would operate by fluctuating between maximum power and long times spent sitting idle. This would result in high investment costs (capacity) and negative effects to the durability of the electrolyzer (fluctuating operation). In addition, there are significant investment costs associated with high-pressure storage tanks for the hydrogen. Using a high power rated battery to smooth the renewable generation peaks mitigates some of these problems. However, it also adds significant complexity and additional investment costs compared to other systems of one technology capable of parallel hydrogen generation and energy storage. There are several options being developed in this regard: Reynard & Girault (2021) propose a vanadium-manganese redox dual-flow battery for this application providing an average energy efficiency of 68 % at a current density of 50 mA/cm² [12]. This highly relevant and innovative approach addresses the problem of renewable integration and storage accompanied by the utilization of earth-abundant elements but comes with the typical high complexity and questionable economic viability of redox flow applications, especially in terms of the requirements for separation units or additional electrolytic cells.

Another option was evaluated by Iranzo & Mulder (2021) in the form of a Ni/Fe hybrid alkaline battery-electrolyzer incorporated with nickel-iron layered double oxides [13]. This allows the combination of a device called a *battolyser* where hydrogen production is no longer seen as a side



Fig. 1. Comparison of CAISO's net demand during one day in 2018 (resembling a duck curve) and 2023 (resembling a canyon curve). Image provided by EPRI.

Table 1

Chemical reactions of alkaline electrolyzer vs. the decoupled water splitting of the ZZE technology.

Electrolyzer					
Overall reaction	$2 \text{H}_2\text{O} \rightarrow 2 \text{H}_2 + \text{O}_2$			$E^\circ_{\text{Cell}} = 1,23 \text{ V}$	(1)
Anode reaction	$4 \text{OH}^- \rightarrow \text{O}_2 + 2 \text{H}_2\text{O} + 4 \text{e}^-$			$E^\circ = 0,40 \text{ V}$	(2)
Cathode reaction	$4 \text{H}_2\text{O} + 4 \text{e}^- \rightarrow 2 \text{H}_2 + 4 \text{OH}^-$			$E^\circ = -0,83 \text{ V}$	(3)
ZZE					
Charging				Discharging	
Overall reaction	$2 \text{Zn(OH)}_4^{2-} \rightarrow 2 \text{Zn(s)} + \text{O}_2 + 4 \text{OH}^- + 2 \text{H}_2\text{O}$	$E^\circ_{\text{Cell}} = 1.60 \text{ V}$	(4)	$2 \text{Zn(s)} + 4 \text{OH}^- + 4 \text{H}_2\text{O} \rightarrow 2 \text{Zn(OH)}_4^{2-} + 2 \text{H}_2$	$E^\circ_{\text{Cell}} = 0.37 \text{ V}$ (6)
Anode reaction	$4 \text{OH}^- \rightarrow \text{O}_2 + 2 \text{H}_2\text{O} + 4 \text{e}^-$	$E^\circ = 0,40 \text{ V}$	(2)	$2 \text{Zn(s)} + 8 \text{OH}^- \rightarrow 2 \text{Zn(OH)}_4^{2-} + 4 \text{e}^-$	$E^\circ = -1.20 \text{ V}$ (7)
Cathode reaction	$2 \text{Zn(OH)}_4^{2-} + 4 \text{e}^- \rightarrow 2 \text{Zn(s)} + 8 \text{OH}^-$	$E^\circ = -1.20 \text{ V}$	(5)	$4 \text{H}_2\text{O} + 4 \text{e}^- \rightarrow 2 \text{H}_2 + 4 \text{OH}^-$	$E^\circ = -0,83 \text{ V}$ (3)

product but instead as a main electrolysis fuel product next to battery function. The authors claim an energy efficiency of 80–90 % with a performance close to the ideal grid-stabilization energy storage [13]. Within this type of cell, the formed Fe and NiOOH electrodes become increasingly active upon fully charging as efficient hydrogen and oxygen evolution occur [14]. This means that hydrogen production within the *battolyser* system is coupled to the fact that the battery needs to be fully charged which can be seen as a limiting factor for this technology.

Zink-Zwischenschritt-Elektrolyseur (ZZE) or Zinc intermediate step electrolyzer in English, is a novel energy storage device being developed by STOFF2. This technology is a combined energy storage and conversion system that decouples the water splitting reaction from hydrogen generation while using zinc to store electrical energy in chemical form. The chemical reactions are presented in Table 1. The overall chemical reaction for traditional alkaline electrolyzer systems follows Eq. (1) and consists of the oxygen evolution reaction (OER) at the anode (2) and the hydrogen evolution reaction (HER) at the cathode (3):

The overall chemical reaction for the ZZE systems also follows Eq. (1) but is decoupled into three operation modes: 1. Charging, 2. Storing, 3. Discharging, illustrated in Fig. 2. A ZZE consists of two electrodes and an alkaline zincate-containing electrolyte. During charging the anode reaction follows the four-electron OER of Eq. (2), while at the cathode the Zn plating reaction takes place following Eq. (5) resulting in the overall charging reaction of Eq. (4). The first step of charging initiates the electrochemical storage of energy in plated zinc. The behavior of zinc plating in alkaline solutions is a well-studied topic. The morphology of the zinc deposits is of particular interest for the ZZE system design and can vary from mossy or spongy, layered, granular, dendrite, and cluster deposit types depending on different parameters such as current density, zincate concentration, ion transport, and temperature [15]. In the second phase, the energy is stored safely in the solid zinc layer.

During the third phase, discharging, the ZZE switches from an energy storage device to an energy conversion device completing the water splitting reaction as the two-electron HER takes place at the cathode following Eq. (3) while the zinc layer is oxidized and dissolves back into the electrolyte forming a zincate complex following Eq. (7). This results in the overall discharging reaction of Eq. (6).

In comparison to traditional (alkaline) electrolyzers, the ZZE has several advantages. Because of the temporal separation of the OER and HER, the formation of an oxygen-hydrogen mixture is eliminated. The ZZE is also free of membranes/separators and therefore does not use harmful perfluorosulfonic acid (PFSA) chemicals or gas-diffusion-layers/flow fields. The ZZE is designed as a low-cost device and is free of noble metal catalysts.

The ZZE is an energy storage and conversion system combined in one device that can operate with a unique asymmetric ratio of high charging power to relatively low discharging power. This can allow ZZE systems to provide balance to renewable based energy grids. At times of the day when there is a high availability of renewable energy and low energy prices, the ZZE can be charged, and energy can be stored within the zinc layer. After that the stored energy can be converted into hydrogen during discharge. Additionally, energy in the form of electricity and

process heat are also discharged from the ZZE system. Therefore, three valuable energy products are produced from a ZZE: hydrogen, heat, electricity.

This represents a significant shift in both energy storage and hydrogen producing technology. Traditional electrolyzers consume electricity while hydrogen is produced. The ZZE process is a way to temporally decouple the consumption of renewable energy from the production of green hydrogen. However, this decoupling makes timing a key factor for the successful operation of ZZE systems. In order to produce totally green hydrogen, charging needs to occur during peaks of renewable energy production and discharging needs to last long enough to span peaks. To accomplish this and successfully operate a ZZE system, a “battery” management system needs to manage the ZZE operation modes via monitoring and controlling of the key parameters such as voltage, current, temperature, as well as an estimation of the SOC. The SOC is a measure of how much energy is held within the system at any point in time. More specifically, SOC is the ratio of the available capacity ($Q(t)$) to the rated charging capacity that can be stored in a system, also referred to as the nominal capacity (Q_n). This can be seen in Eq. (8). The SOC is a critical value for an effective operation of the “battery”, or more accurately, ZZE management system.

$$\text{SOC}(t) = \frac{Q(t)}{Q_n} \quad (8)$$

An inaccurate estimation of the SOC of a system can significantly affect the operation of the ZZE. Especially in extreme cases of overcharging or discharging. When a ZZE is operated in a charging mode for too long, the electrolyte can become Zn-depleted and could switch from decoupled water splitting into regular coupled water splitting, meaning OER and HER would occur simultaneously. Due to the lack of a membrane, this operation mode could result in oxygen and hydrogen mixing and must be prevented. When the discharging mode is not stopped when the SOC reaches zero (or before), the polarity in the cell could reverse leading the reactions to occur at the incorrect electrode. This could alter the catalytic surfaces and ultimately lead to a reduction in performance. Both extreme cases of undesired side reactions result in reduced efficiencies and ultimately cause early system failure. Indirectly, this could also lead to renewable energy that could otherwise have been stored within the system being curtailed.

The SOC can be challenging to measure directly. There are different estimation techniques to predict it. These can be either performed offline or online. Offline techniques can only be performed when interrupting the system and involve the consideration of the internal physics such as the resistance, capacitance, or voltage characteristics [16]. Due to the high requirements for the experimental conditions and time- and resource-intensive requirements, these offline approaches are more suitable for academic research or as a cross-check for other SOC prediction methods rather than for applied operations. Therefore, other more pragmatic solutions are required in order to accomplish rapid technology development, industrial production, and ultimately commercial deployment.

Online techniques include look-up table, ampere-hour integral,

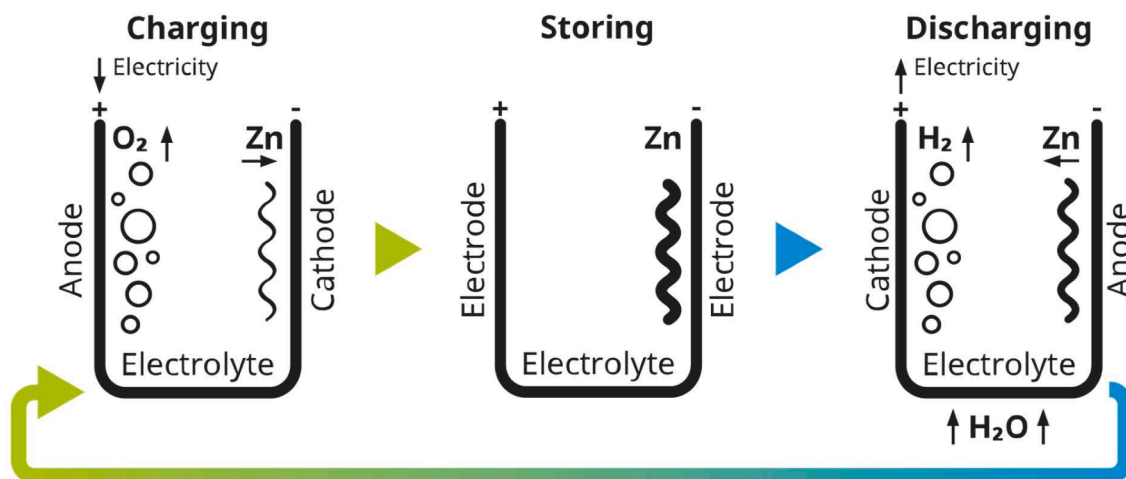


Fig. 2. Scheme of the distinct operation modes of a ZZE: Charging – Storing – Discharging.

filter-based, observer-based, and data-driven methods [17]. The look-up table and ampere-hour integral are relatively straightforward methods but have issues with low accuracy and poor robustness in dealing with sensor errors [17]. Filter and observer-based methods have relatively higher estimation accuracy, self-corrective ability, and better robustness in dealing with noise compared to look-up table and ampere-hour integral methods [18]. However, they also require a large number of time-consuming and complicated tests to build a well-parameterized model of the studied system [19].

In this work, various open-source ML algorithms are used to train models capable of predicting the SOC of a lab scale ZZE system. The results are analyzed based on the effectiveness of each applied model. Ultimately, this will help STOFF2 determine the practicability of each ML framework as a basis for future ZZE management systems responsible for avoiding extreme operation cases that can result in damage to the ZZE system.

Materials and methods

Recent advancements in AI technology have highlighted the advantages of data-driven methods compared to other estimation techniques [20]. Data-driven methods only require measurement data for accurate and robust SOC estimation rather than detailed formulas or models [21]. The relationship between battery parameters and the SOC is non-linear and the key parameters that affect SOC include voltage, current, and temperature among others. In data-driven methods these measurable parameters become the features that ML or other frameworks use to predict a system's SOC. This is accomplished by recording cycle data and constructing a dataset consisting of the measured battery parameters (features) and the resulting SOC of a given storage system

(output). This dataset is then separated into training and test sets. The selected framework uses the training set to map the relationship between the features and the output. The test set is then used to measure the accuracy (or in this case error) of the newly trained model by passing it the feature data of the test set and predicting the output. This predicted output is compared to the observed output by calculating—in this case—the mean absolute error (MAE) between the two values.

For this project a 12 Wh single cell monopolar ZZE reactor (Fig. 3) was designed and constructed by STOFF2. This cell was cycled 81 times at STOFF2's lab in Berlin. ZZE systems, unlike traditional electrolysis technologies that are classified with a power rating, are classified by the total hydrogen capacity of one typical daily cycle (12 Wh based on the higher heating value of hydrogen is roughly equal to 0.04 kg hydrogen). The feature data for these cycles were collected using a Keysight Truevolt 34465A Tabletop Multimeter and a Keysight E3648A Dual Output DC Power Supply. The cell was also cycled 66 times at the Technologiezentrum Energie at HAW Landshut in order to provide third-party verification of the results. Data for these 66 cycles were collected using a Biologic BCS-815 (SN 0361) battery cycler using BT-Lab v1.72 software. The temperature was set to a constant level throughout both the charging and discharging cycles using external electrical heating elements for the cycles conducted at STOFF2. Whereas the Biologic BCS-815 battery cycler used at HAW Landshut has an internal temperature control. The test cycles were operated at a constant current for the charging and discharging cycles. These current values differ due to the designed asymmetry of the ZZE. The voltage response of the cell was recorded by the aforementioned external battery management systems.

During each charging cycle the cell was charged to 50 % of the total nominal capacity of the device in order to allow for an accelerated

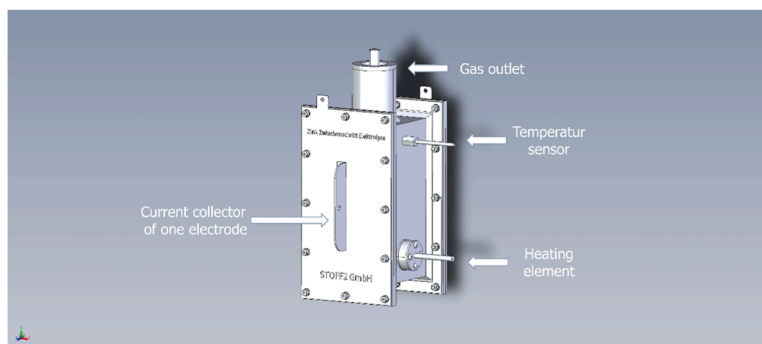


Fig. 3. Schematic drawing of the 12 Wh ZZE test cell.

testing procedure. However, for the sake of simplicity, after each charging cycle the SOC was assumed to be at 100 %. As such, the dependent variable for this project was the SOC of the discharge cycle from 100 % to 0 %, calculated via ampere hour counting. Observations were recorded every 60 s. The ampere hours discharged at each observation were cumulatively summed and divided by the total ampere hours discharged over each cycle. This effectively estimates the state of discharge (SOD) of each cycle. The SOC was calculated by subtracting this SOD value from 1 ($1 - \text{SOD} = \text{SOC}$). This also means that the model is implicitly predicting the partial energy discharged divided by the total energy discharged at each observation. The trigger for the end of each discharge cycle was when the voltage reached 0.05 V for the cycles conducted at STOFF2 and 0 V for the cycles conducted at HAW Land-shut. The features were the discharge voltage, current, and temperature as well as the electrolyte concentration and number of observations recorded for each cycle. Cycles show varied length depending on the charging/dischARGE current and also the stochastic nature of each cycle. The cycles were then randomized and split into 80 % training and 20 % testing datasets of 117 and 30 cycles respectively.

Theory/calculation

The intent of this research is to predict the SOC over an entire discharge cycle. This means that the intended prediction is the entire discharge SOC sequence not just a single value. Models that use sequences to predict subsequent sequences are known as sequence-to-sequence (seq2seq) models. Based on literature reviewed prior to building the applied models, it was hypothesized that a Long Short-Term Memory (LSTM) recurrent neural network (RNN) seq2seq model would be best suited to this project. Theory suggests that an LSTM based recurrent neural network would be a good fit to predict the SOC discharge sequence because LSTM based models are more equipped to capture long term dependencies in data due to the LSTM cell's efficient, gradient based method which truncates the gradient where it does no harm [22]. These models learn to bridge time lags over 1000 timesteps long by enforcing constant error flow through constant error carousels within special units [22]. The special memory units store a summary of the previous inputs, and a gating mechanism controls the information flow between input (i_t), output (o_t), and cell memory (c_t) [21]. This prevents back-propagated errors from vanishing or exploding and is more suited to accurately predicting long sequences and can be seen in Fig. 4 [23].

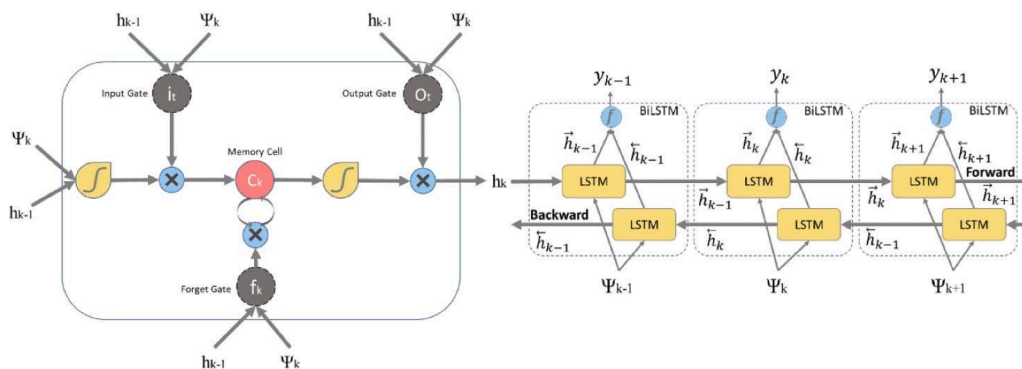


Fig. 4. Standard LSTM layer (left): Ψ_k and h_{k-1} are the input data layer at the current time-step k and hidden layer at the previous time step $k - 1$, respectively. i_k is the input which regulates how much of past information should be passed to the memory cell. o_k is the output which controls how much of the information computed in the memory cell will be included in the output (h_k). f_k is the forget gate which selects how much of the previous information in c_{k-1} should pass ahead to the next time. And c_k is the memory cell which stores and transfers information from previous and current states that is to be used in future. Bi-Directional LSTM layer (right): Composed of a forward and backward “unidirectional” LSTM layer, where the forward part is fed with temporal input data Ψ starting from the earliest time-step Ψ_{k-n} to the current time step Ψ_k . The backward part is fed in reverse order from Ψ_k to Ψ_{k-n} . From the forward LSTM part Ψ_k and $\tilde{h}_{k-1}$ are respectively the input data at current time-step k and hidden layer at the previous time step $k - 1$, wherefrom the backward LSTM the inputs are Ψ_k and $\tilde{h}_{k-1}$. The BiLSTM output, y_k , are given by the combination of both forward and backward LSTMs, $\tilde{h}_k$ and $\tilde{h}_k$, through chosen function f . Provided by Vidal et al. (2020).

Chemali et al., (2017) trained an LSTM RNN model on datasets of a Panasonic 18650 battery cell with a lithium nickel cobalt aluminum oxide (LiNiCoAlO₂ or NCA) chemistry at various ambient temperatures [24]. This battery cell was exposed to 10 different drive cycles consisting of a mix of the Highway Fuel Economy Test Cycle (HWFET), Urban Dynamometer Driving Schedule (UDDS), LA92 Unified Driving Cycle, and Supplemental FTP (US06) driving schedules for an electric Ford F150 truck [24]. Eight or nine of these cycles became the network's training data with one or two being set aside as a validation dataset, and one cycle used as a testing set [24]. The LSTM-RNN Chemali et al., (2017) developed achieved an MAE of 0.573 % at a fixed ambient temperature and a MAE of 1.606 % on a dataset with ambient temperature increasing from 10 °C to 25 °C.

In order to develop more advanced neural network models, the ZZE dataset described in the previous section needed to be reshaped into a 3-D tensor comprised of the samples, timesteps, and features. In this case there are 146 samples (cycles) with 1339 timesteps (total number of observations for the longest cycle), and five features (voltage, current, temperature, electrolyte concentration, and observations). Cycles with less than 1339 observations were masked to ensure that no data was lost. This 3-D tensor data was also split into training and testing sets of 117 and 30 cycles, respectively.

Results

Due to the nature of the chemical reactions, the ZZE charge-discharge-cycles show moderate variation from cycle to cycle, especially in comparison to other battery technologies. Causes of this relatively higher cycle to cycle variation include the recognized variation in the Zn layer formation during charging, oxide transformation on the catalytically active surface of the bifunctional gas electrode (at the + pol where both oxygen and hydrogen gas are temporally decoupled and produced), and effects determined by the specific reactor design that still need to be further identified to optimize ZZE operation. Some of this can be seen in the voltage output for the cycles used in this project (Fig. 5). This will also be discussed in more detail in the following section.

In order to compare the different trained ML models, a baseline is needed to contextualize the results. The baseline chosen to compare the applied models to was an ordinary least squares (OLS) linear regression. If the tested models do not outperform a linear regression, then the added complexity of the model would not provide enough benefit to be

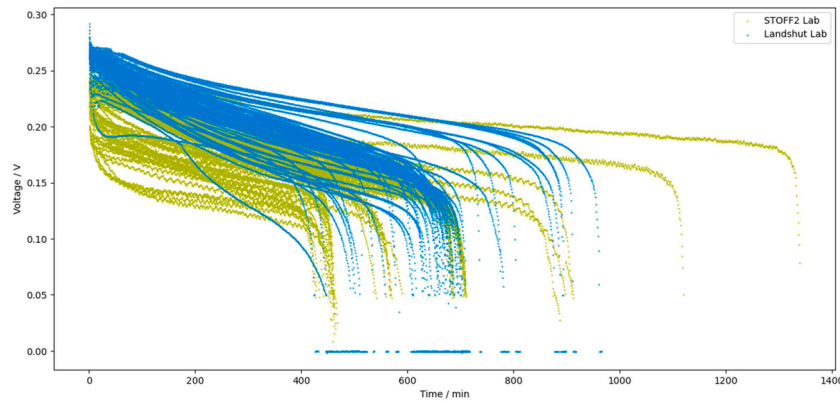


Fig. 5. Discharge voltage response curves for all cycles experimentally recorded with the 12 Wh ZZE test cell. Note that the temperature control at HAW Landshut is more precise at maintaining a constant temperature. Hence, the voltage responses curves from STOFF2 show higher fluctuation.

of interest to the SOC prediction.

The linear regression used was the ordinary least squares linear regression from the Scikit Learn open-source ML library [25]. Like all linear regressions this model minimizes the residual sum of squares between the observed targets in the dataset and the targets predicted by linear approximation [25]. Fitting this regression to the training dataset, the feature data (voltage, current, temperature, electrolyte concentration, observations) of the test dataset was used to predict the SOC. This predicted SOC was then compared to the observed SOC value via the mean absolute error (MAE) between the two sequences. This value is the baseline used to compare the rest of the models to. The MAE between the linear regression predicted SOC and the actual observed data was 7.94 %.

The first method tested was a supervised learning method called a support vector machine (SVM) regression also from the scikit learn library²⁵. Generally, many regression problems cannot be linearly regressed in the space of the inputs; the advantages of SVM are that they can map feature vectors to a higher-dimensional space that can be linearly separated [17]. This means that SVMs have the advantage of good regression ability under small sample sizes [17]. With some trial and error, the optimal configuration of the SVM regression was found to be with a regularization parameter equal to 10 and a kernel coefficient value of 0.1. This resulted in a MAE of 6.86 %, outperforming the baseline linear regression.

The next method tested was ensemble learning. The main objective of ensemble learning is to reduce the risk of choosing a single learning algorithm with a poor performance and to improve model performance by using an intelligent ensemble of several individual algorithms [26]. The specific ensemble learning method used in this work was a random forest (RF) regressor, also from the scikit learn library [25]. It is a meta estimator that fits a number of classifying decision trees on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control for over-fitting [25]. Again, through trial and error the regression was optimized with 100 trees in the forest and no maximum depth of each tree. This resulted in a MAE of 4.60 % which significantly outperforms the OLS regression.

As mentioned above, in order to build artificial neural network (ANN) models, the data was reshaped into a 3-D tensor. The Keras open-source library [27]—using TensorFlow as backend and the sequential class to add layers together—was used to build the ANN models tested.

The first neural network model tested was a standard ANN (no LSTM layers). This model was found to perform optimally when configured with two fully connected layers with 12 and eight neurons respectively. In order to predict a sequence as output these fully connected layers are followed by a time distributed fully connected layer with one node. The time distributed wrapper function of the single neuron applies the output of the preceding layer to each temporal slice. Ultimately, this will

output a single masked sequence with the same number of observations as input (1339). The Adam optimizer, a learning rate of 0.001, and a batch size of 32 were used. The model was compiled with the loss function set to the MAE. To prevent over training 10 % of the training data was set aside to use as a validation set. Early stopping was set to monitor the loss function of this validation set; when the validation loss (MAE) did not improve for 10 consecutive epochs, the model will finish training. Due to the stochastic nature of neural networks this model was run three times, and the average MAE was taken in order to present a representative result. The model ran for a mean of 69 epochs before early stopping was enacted and the average MAE across runs was 4.66 % (the average validation MAE at the end of model training was 2.4 % across these three runs).

The LSTM model tested was configured in much the same way as the standard neural network above; again, the Adam optimizer, a learning rate of 0.001, and a batch size of 32 were all used. The model was also compiled with the loss function set to the MAE and the same early stopping was enacted. The optimum configuration of the LSTM model was found to be a network with two bidirectional LSTM layers with two and 20 hidden cells respectively. The difference between a normal LSTM layer and a bidirectional LSTM is that the standard LSTM can only access the previous input of each specific time step [21]. After adding the bidirectional recurrent structure, it can process data in two directions, forward and backward [21]. Thus, each bidirectional LSTM hidden layer can jointly learn past (forward) and future (backward) dependencies at a specific time step [21]. This can be seen in the right side of Fig. 4 [23]. Again, the model was run three times with a mean length of 106 epochs and an average MAE across runs of 3.21 % (the average validation MAE at the end of model training was 2.0 % across these three runs). Fig. 6 shows the predicted (magenta) and actual (blue) SOC values of the test set of one of the three times the model was run.

Table 2 highlights how each model performed relative to the linear baseline.

Discussion

The results from these models confirm the initial hypothesis that an LSTM based RNN would be a good fit to predict the SOC discharge sequence for ZZE systems. However, the results also showed that a random forest regressor and standard neural network are capable of relatively accurate predictions. When one considers the efficiency of each of these algorithms, the 1.45 % and 1.39 % lower MAE that the LSTM model was able to achieve may not be worth the added time and compute necessary to run the more complex model. Running locally on a 1.8 GHz Dual-Core Intel Core i5 processor the LSTM regressions took an average of six min and 38 s to train. Whereas the standard ANN and random forest regressor on average took 12 and 19 s to train

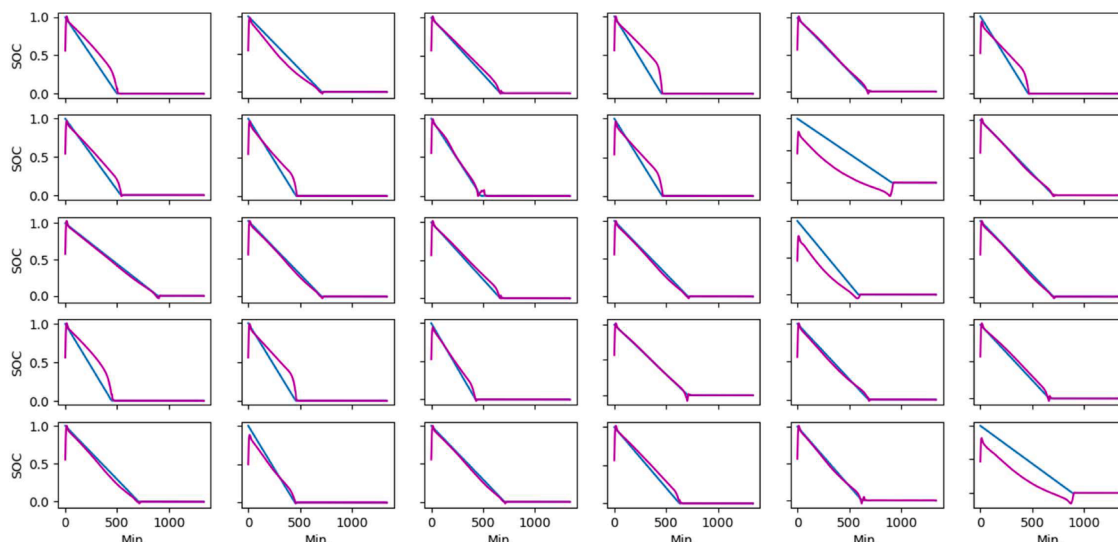


Fig. 6. Output SOC discharge curves for all 30 cycles of the test set of one selected test run through the LSTM based neural network.

Table 2

Comparison of the MAE of all tested regression models.

Model	OLS - Baseline	SVM	RF	ANN	LSTM
MAE	7.94 %	6.86 %	4.60 %	4.66 %	3.21 %

respectively. Considering that in practice ZZE charging decisions will need to be made in real time, efficiency may be another factor to consider. Another way to further decrease the time-consumption of these models would be to run them on a central processing unit (CPU) with more processing power, or graphical processing unit (GPU).

It is also important to recognize that the tested LSTM network was not able to replicate the MAE of LSTM models used to predict the SOC of commercial Li-ion batteries. Li-ion batteries are a mature technology which have been subject to years of testing and optimization. However, there are also fundamental differences between the charging and discharging processes of Li-ion batteries and ZZE. A ZZE cycle involves the three-phase boundary of electrode-electrolyte-gas interfaces at the bi-functional gas electrode. Combined with the stochastic nature of the Zn plating/deplating reactions, this results in higher variability in the voltage response. The oxide morphology of this bi-functional electrode surface is constantly in flux. The specific species of oxides that are present on the catalytic surface vary with each operation, not only from one cycle to another but also from the charging to subsequent discharging of each cycle.

In contrast, conditioning of a Li-ion battery is conducted via some initial cycles of slow/gentle charging and discharging before it is first used. This forms the so-called solid-electrolyte interphase (SEI), leading to an initial interface layer from the reduction of the electrolyte. This layer chemically rearranges during conditioning (repeated and defined cell aging) and stays more or less stable within the lifetime of the battery [28].

The SEI layer allows the conduction of lithium ions and furthermore reduces the contact of electrolyte with the electrode, thus reducing electrolyte decomposition. A robust SEI layer is required for stable battery operating conditions. On the other hand, the presence of the SEI layer increases the impedance in lithium-ion batteries, thus leading to capacity losses in the battery that are not present in ZZE operation [29].

There is no SEI that needs to be established on ZZE electrodes. The zinc layer is repeatedly created and dissolved upon charging and discharging of the cell and its thickness and porosity is defined by the applied voltage, current and charging time. This may partially explain why similar LSTM models are able to predict lithium-ion batteries' SOC

with a lower MAE. However, the comparison is also difficult when comparing a highly engineered and mature system to a first-of-its kind laboratory test cell.

Another possible explanation could be the relatively limited variability of the dataset used in this project. The test cell was cycled with the intent of further understanding its function. As discussed above, there is an intrinsic variability within each ZZE process. As such, it was cycled with the intent of specifically maintaining cycle to cycle repeatability in order to better understand the variation caused by each process. There was still enough variation present throughout the entire dataset for the tested models to learn how to map the selected features to the output (SOC) but adding more varied cycle data could also improve the performance of the tested models. This lack of data is in part due to the accompanying circumstances of the rapid ZZE development and limited financial, personnel, and equipment available to STOFF2.

The reality of the energy transition necessitates the rapid advancement of technical solutions for the storage of renewable energy and the production of green hydrogen. Modern and agile research approaches have to be undertaken. Open-source ML techniques are effective pattern recognizers and can help human agents investigate complex mechanisms, even though the relevant chemical characteristics vary from situation to situation, as is the case in ZZE operation [30]. This provides an enormous head start for the development of key components of complex technical systems such the management system of future commercial ZZE systems. The use of AI can help new technological innovations deal with the complexity of operating within an increasingly intricate electricity market dominated by renewable energy. This will ultimately serve to decrease the time it will take to deploy innovative solutions—such as the ZZE—in line with the broader goal of working towards a rapid energy transition.

Conclusion

This work presents the novel technology of *Zink-Zwischenschritt-Elektrolyseur* (ZZE), or Zinc Intermediate step Electrolyzer in English. This innovative technology decouples the water splitting reaction into two parts and therefore combines the features of energy storage (electrical energy is stored in chemical form as solid zinc) and energy conversion (de-plating of zinc results in hydrogen gas production) in one device. Due to the nature of the ZZE charging and discharging processes, an accurate state-of-charge prediction remains challenging.

We evaluated various machine learning models including ordinary least squares, support vector machine, random forest, standard artificial

neural network, and long short-term memory network regressions for their effectiveness. The long short-term memory approach performed the best, with an average mean absolute error of 3.21 %.

We show that the use of machine learning tools can help predict the state of charge which is a precursor to the full development of a ZZE management system. Building on these results STOFF2 will continue to pursue ZZE development and the work necessary to better understand the complex underlying operational processes of the technology. Next steps include repeating similar experiments with larger and industry relevant ZZE systems. These subsequent experiments should also include more intracycle feature variation, which conveniently mirrors how the technology is likely to be cycled in practical applications. In order to validate the results from this initial data-driven machine learning approach, complementary state of charge prediction methods based on internal physics and electrochemical reactions will also be studied in future.

CRedit authorship contribution statement

Daniel Vila: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Visualization, Writing – original draft, Writing – review & editing. **Elisabeth Hornberger:** Investigation, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Christina Toigo:** Project administration, Supervision, Writing – original draft, Writing – review & editing, Data curation, Funding acquisition, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

Funding: This work was supported by the German Federal Ministry of Education (grant number 01LY2111A) and the German Federal Ministry of Economics and Climate Action (grant number 03EI3092A).

Special thanks to Drs. Thomas and James McDermott for their continued mentorship and support in the ideation process of this work.

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